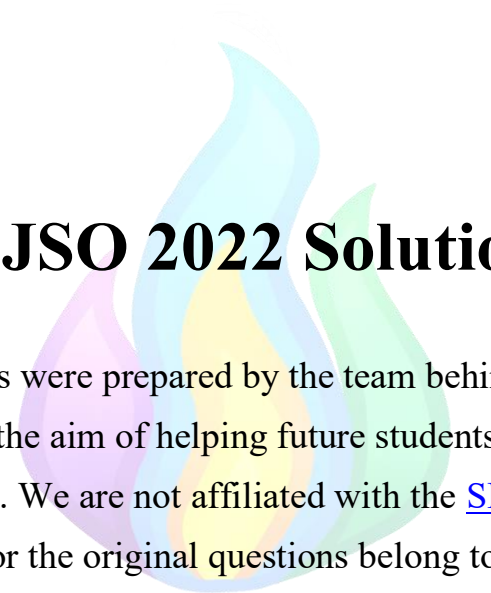




SLJSO 2022 Solutions

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Question No.	Field	References
<u>1</u>	Cell Biology	<u>Prokaryotic vs. Eukaryotic Cells</u>
<u>2</u>	Cell Biology	<u>Cell Structure and Organelles</u>
<u>3</u>	Cell Biology	<u>Cell Structure and Organelles</u>
<u>4</u>	Cell Biology	<u>Biomolecules</u>
<u>5</u>	Human Biology	<u>Heart</u>
<u>6</u>	Ecology	<u>Ecological Pyramids</u>
<u>7</u>	Molecular Biology	<u>Structure of DNA and RNA</u>
<u>8</u>	Cell Biology	<u>Passive Transport</u> <u>Cell Structure and Organelles</u>
<u>9</u>	Botany	<u>Transport in Plants</u>
<u>10</u>	Botany	<u>Transport in Plants</u>
<u>11</u>	Ecology	<u>Nitrogen Cycle</u>
<u>12</u>	Human Biology	<u>Circulatory System</u>
<u>13</u>	Human Biology	<u>Circulatory System</u>
<u>14</u>	Cell Biology	<u>Mitosis and Meiosis</u>
<u>15</u>	Molecular Biology	<u>Gene Expression</u>
<u>16</u>	Chemical Structures	<u>Subatomic Particles</u>
<u>17</u>	Chemical Structures	<u>Electron Configurations</u>
<u>18</u>	Chemical Structures	<u>Periodic Table Trends</u>
<u>19</u>	Stoichiometry	<u>Mole</u>
<u>20</u>	Stoichiometry	<u>Reaction Stoichiometry</u>
<u>21</u>	Stoichiometry	<u>Mass Percentage</u>
<u>22</u>	Thermochemistry	<u>Temperature Scales</u>
<u>23</u>	Stoichiometry	<u>Reaction Stoichiometry</u>
<u>24</u>	Stoichiometry	<u>Concentration</u> <u>Reaction Stoichiometry</u>
<u>25</u>	Stoichiometry	<u>Reaction Stoichiometry</u>

<u>26</u>	Stoichiometry	<u>Reaction Stoichiometry</u>
<u>27</u>	Chemical Structures	<u>Intermolecular Forces</u>
<u>28</u>	Stoichiometry	<u>Reaction Stoichiometry</u>
<u>29</u>	Other Chemistry Topics	<u>Properties of Common Compounds</u>
<u>30</u>	Stoichiometry	<u>Reaction Yield</u>
<u>31</u>	Mechanics	<u>Density</u>
<u>32</u>	Electromagnetism	<u>Transformers</u>
<u>33</u>	Electromagnetism	<u>Series and Parallel Circuits</u> <u>Kirchhof's Laws</u>
<u>34</u>	Light	<u>Mirrors</u>
<u>35</u>	Thermodynamics	<u>Specific Heat Capacity</u>
<u>36</u>	Mechanics	<u>Torque</u>
<u>37</u>	Mechanics	<u>Motion Graphs</u>
<u>38</u>	Thermodynamics	<u>Ideal Gas Law</u>
<u>39</u>	Light	<u>Refraction</u>
<u>40</u>	Mechanics	<u>Energy Conservation</u>
<u>41</u>	Mechanics	<u>Newton's Laws</u>
<u>42</u>	Mechanics	<u>Energy and Work</u> <u>Power</u>
<u>43</u>	Electromagnetism	<u>Lenz's Law</u>
<u>44</u>	Mechanics	<u>Buoyancy</u>
<u>45</u>	Light	<u>Lens Formula</u>
<u>46</u>	Algebra	
<u>47</u>	Algebra	
<u>48</u>	Algebra	
<u>49</u>	Algebra	
<u>50</u>	Area	

Question 1

This problem tackles metabolic processes, specifically cellular respiration.

Option (A) describes glycolysis, which has a net gain of 2 ATP molecules. **Option (B)** describes cellular respiration as a whole, which produces 32-38 ATP (depending on the cell). Option (C) describes lactic acid fermentation, and option (D) describes alcoholic fermentation, both of which produce only 2 molecules of ATP per molecule of glucose. By comparing the numbers, we find that **option (B)** (Glucose $\rightarrow$ CO₂) is the correct answer.

Correct Answer: B

Question 2

This problem focuses on prokaryotic cell structure and function.

Option (A) states that prokaryotic cells contain membrane-bound organelles, which is incorrect.

Options (B) and (C) both describe things that are found in both eukaryotic and prokaryotic cells.

Option (D) is the correct answer because nitrogen-fixing bacteria are the only organisms with the ability to fix atmospheric nitrogen, and all bacteria are prokaryotic.

Therefore, **option (D)** is the correct answer.

Correct Answer: D

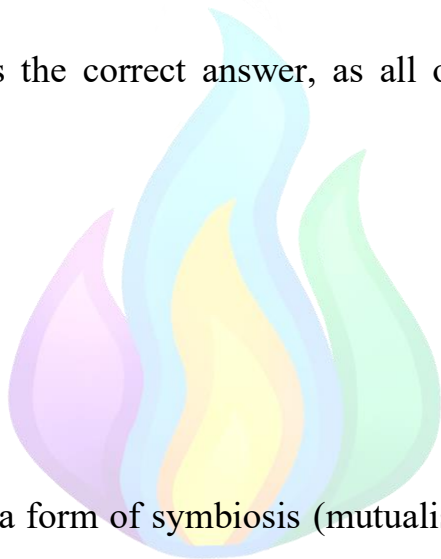
Question 3

Benedict's solution is used to detect the presence of reducing sugars by turning red, like glucose, fructose, galactose, lactose, and maltose.

Option I is a glucose solution—which is a reducing sugar, so it will turn red. In Option II, HCl hydrolyzes (breaks apart with water) sucrose into glucose and fructose—both of which are reducing sugars—causing the solution to turn red. In option III, starch is broken down by amylase into maltose and glucose—both of which are reducing sugars—turning the solution red as a result.

Therefore, **Option (D)** is the correct answer, as all of the solutions form a red precipitate.

Correct Answer: D



Question 4

Option (A) is considered a form of symbiosis (mutualism) because the fungi helps decompose and transform nutrients to make them available for the plants. The plants give back to fungi by providing it with sugars from photosynthesis. Option (B) is considered symbiotic (parasitism). The cuscuta plant is taking nutrients from the plant without giving anything back. However, symbiosis can be either beneficial, harmful, or neutral—as long as it involves a close relationship between two or more organisms. Option (C) is considered symbiotic (mutualism). The bacteria fix the nitrogen into a usable form for the plant, while the plant provides the bacteria with sugars produced by photosynthesis. **Option (D)** is not considered symbiotic because the lichen is interacting with the fungi and the moss on the bark, not the bark itself.

Correct Answer: D

Question 5

Photosynthesis is an energy absorbing process (it takes energy from photons and splits them to get electrons).

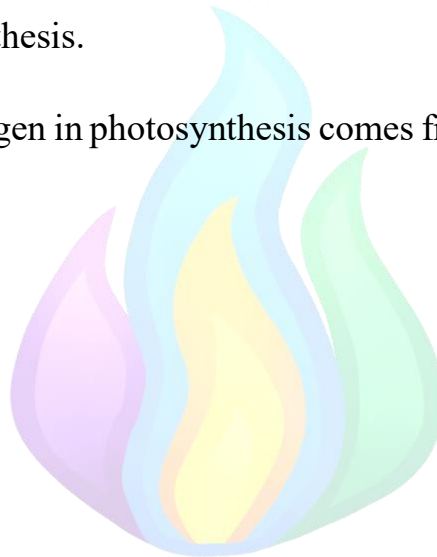
Option (A) is false for that reason.

Option (B) is correct, chlorophyll is best at absorbing red and blue light, as seen in an absorption spectra graph.

Option (C) is correct, there are other pigments like carotenoids and xanthophylls that can be a part of photosynthesis.

Option (D) is correct, oxygen in photosynthesis comes from the splitting (photolysis) of water.

Correct Answer: A



Question 6

Option (A): Vitamin A is essential when it comes to vision, skin, and immune health.

Option (B): Vitamin B is essential for energy metabolism and red blood cell formation.

Option (C): Vitamin C is essential for collagen synthesis and antioxidation.

Therefore, **option (D)**, vitamin K, is the correct answer. It is essential for the synthesis of clotting factors in the liver (e.g., prothrombin).

Correct Answer: D

Question 7

Option (A) is incorrect, because evaporation of water from the skin (sweat) is a cooling mechanism, which occurs due to an increase in body temperature, not a decrease.

Option (B) is correct. Metabolism expends energy, releasing heat. Hormones like thyroxine and epinephrine regulate metabolism, which in turn regulates body temperature. If body temperature is high, these hormones will lower metabolic rate to lower body temperature.

Option (C) is incorrect, because hair erector muscles in the skin contract when body temperature is low, known as goosebumps. Hair erector muscles actually relax when body temperature is high to easily release heat.

Option (D) is incorrect, because blood vessel dilation (vasodilation) actually occurs with high body heat to dissipate heat. Constriction of blood vessels (vasoconstriction) occurs with low body heat to retain heat.

Therefore, **option (B)** is the correct answer.

Correct Answer: B

Question 8

This question is about genetic variation.

Option (A): Independent segregation does in fact increase genetic variation, as alleles can segregate independently of one another, increasing randomness. **Option (B):** Linkage is the correct option. Linked genes are inherited together, decreasing the amount of variation available. Option (C): Crossing over increases genetic variation by allowing for the swapping of alleles. Option (D): Incomplete dominance increases genetic variation, as it leads to a wider range of phenotypes.

Therefore, **option (B)** is the correct answer.

Correct Answer: B

Question 9

This is a genetic cross question. Right away, we know that the plant with the white flower and yellow seeds must have the genotype $rrgg$. White flowers and yellow seeds are recessive, so in order to exhibit those traits, the individual must be homozygous recessive. We can also generate all the possible combinations of the offspring's genotypes: red flowers with green seeds could be one of the following: $RRGG$, $RrGg$, $RRGg$, or $RrGG$. The white flowers with green seeds could be $rrGG$ or $rrGg$.

Testing each one in a genetic cross leads to the correct answer being **option (B)**, $RrGG$. When it crosses with $rrgg$, it produces $RrGg$ (producing red flowers with green seeds) and $rrGg$ (producing white flowers with green seeds).

Correct Answer: B

Question 10

Option (A) is true. The intestine is home to trillions of microorganisms aiding in enzyme production and digestion.

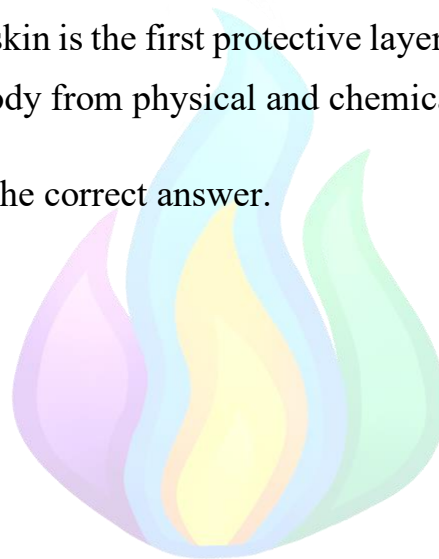
Option (B) is true, because pathogens can release enzymes to aid in invasion and break down the host cell.

Option (C) is false, because endotoxins are heat stable. Exotoxins are the unstable ones.

Option (D) is true, as the skin is the first protective layer against pathogens and other dangers, protecting the body from physical and chemical threats.

Therefore, **option (C)** is the correct answer.

Correct Answer: C



Question 11

Option (A) is true. ATP is a nucleotide (it consists of an adenine base attached to a ribose sugar with 3 phosphate groups).

Option (B) is also true. ATP is produced during photosynthesis.

Option (C) is false. The most ATP per molecule of glucose is actually produced during the ETC (electron transport chain).

Option (D) is true. ATP is used in muscle contraction.

Therefore, **option (C)** is the correct answer.

Correct Answer: C

Question 12

In this question, we are looking for a passive process. Passive processes don't require energy input—they are spontaneous.

Option (A) is an active process, using energy to pull water up. Option (B) is also an active process. The movement of sucrose from mesophyll cells into sieve tubes requires energy. **Option (C)** is a passive process. Sucrose molecules move passively from sieve tube to sieve tube based on a pressure gradient. Option (D) is an active process, as ion movement in the proximal convoluted tubule requires energy.

Therefore, **option (C)** is the correct answer.

Correct Answer: C

Question 13

Option (A) is incorrect, because the skin is actually the largest organ of the body.

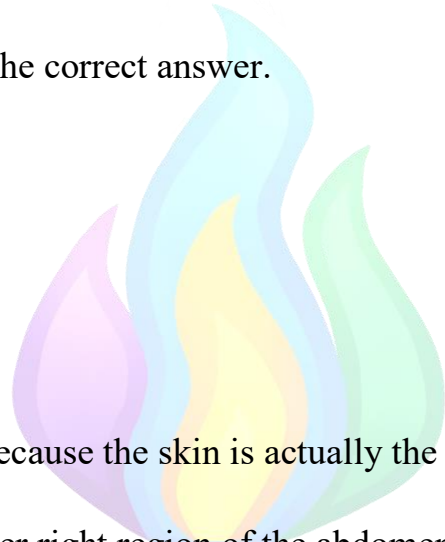
Option (B) lies in the upper right region of the abdomen, so it is incorrect.

Option (C) is incorrect, because bone marrow synthesizes hemoglobin, not the liver.

Option (D) is correct because it produces bile, which plays a role in the digestion of food.

Therefore, **option (D)** is the correct answer.

Correct Answer: D



Question 14

Option (A) is true, because skeletons assist in locomotion.

Option (B) is true, because annelids do in fact have a hydrostatic skeleton.

Option (C) is true, as organisms are typically confined to their exoskeleton and grow into them.

Option (D) is false. Some invertebrates have endoskeletons, such as echinoderms and cephalopods.

Therefore, **option (D)** is the correct answer.

Correct Answer: D

Question 15

Option (A) is false. Most plant viruses contain RNA.

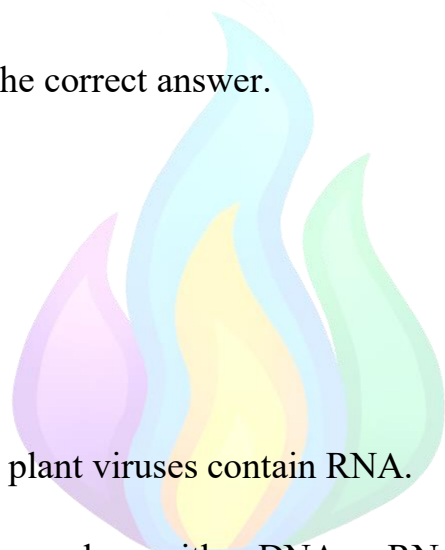
Option (B) is true. Viruses can have either DNA or RNA.

Option (C) is true. Viruses can be used as vectors to deliver genetic material, which is helpful in recombinant technology.

Option (D) is true. All viruses are obligate parasites, as they require a host to replicate and cannot survive outside of a cell.

Therefore, **option (A)** is the correct answer.

Correct Answer: A



Question 16

Theory of the question

- **Isotopes** are atoms of the same element that have the same number of protons but different numbers of neutrons in the nuclei.
- **Mass Number** = Protons + Neutrons

	Protons	Neutrons
P	18	19
Q	16	19
R	18	18
S	17	20

Solution

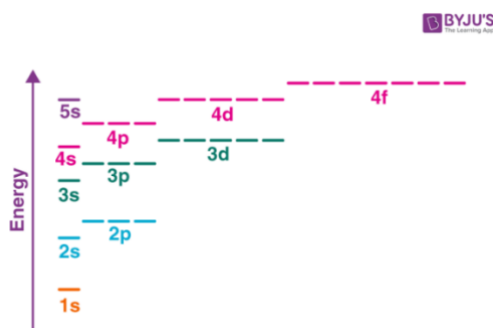
- Atom P has 18 protons.
- Atom R also has 18 protons, so **R** is an isotope of P.
- Mass number of **P** = $18 + 19 = 37$
- Check other atoms:
 - **Q** = $16 + 19 = 35$
 - **R** = $18 + 18 = 36$
 - **S** = $17 + 20 = 37$
- Atom **S** has the same mass number as

Correct Answer: A

Question 17

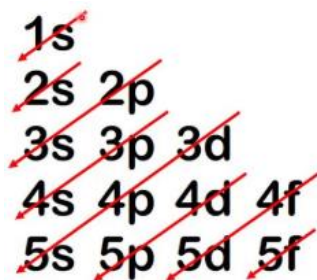
Theory of the question

Aufbau principle:



The Aufbau principle states that electrons fill lower-energy atomic orbitals before filling higher-energy ones.

s 2 e⁻
p 6 e⁻
d 10 e⁻
f 14 e⁻



Hund's rule: each orbital gets one electron before pairing.

Solution:

- Phosphorus has **15 electrons**.
- The electron configuration: $1s^2 2s^2 2p^6 3s^2 3p^3$
 - o 1s can hold **2** electrons $\rightarrow 1s^2$
 - o 2s can hold **2** electrons $\rightarrow 2s^2$
 - o 2p can hold **6** electrons $\rightarrow 2p^6$
 - o 3s can hold **2** electrons $\rightarrow 3s^2$
 - o 3p can hold **3** electrons $\rightarrow 3p^3$
- The **1s, 2s, and 3s** orbitals should be **fully filled** (paired electrons).
- The **2p** orbital should be **fully filled** (paired electrons).
- The **3p** orbital should have **three unpaired electrons** (Hund's rule).

Correct Answer: D

Question 18

Theory of the problem:

- Reactivity of alkali metals increases down the group.
- Atomic radius increases down a group and decreases across a period from left to right.
- Metallic character decreases across a period from left to right.
- Softer metals are found lower in the group because the metallic bonds become weaker.

Solution:

(A) Na is more reactive than K → False

- We know that K (potassium) is below Na (sodium) in Group 1; K is more reactive than Na, not the other way around.

(B) Atomic radius of Mg is larger than that of K → False

- K is in Group 1, and Mg is in Group 2, meaning K is to the left of Mg.
- Potassium (K) has a larger atomic radius than magnesium (Mg).

(C) Li has more metallic character than Be → True

- Li (lithium) is in Group 1, while Be (beryllium) is in Group 2.
- Group 1 elements have higher metallic character than Group 2 elements.
- So, Li is more metallic than Be.

(D) Li is a softer metal than Na → False

- Na is below Li in Group 1, meaning Na is softer than Li.

Correct Answer: C

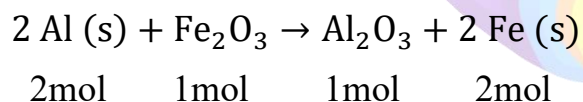
Question 19

- The problem says that one mole of H₂O molecules has a mass of 180 g.
- Instead of using Avogadro's number (6.022×10^{23} molecules/mol), we define a mole as the number of petals in the given flower.
- We see that the flower has 8 petals, so 1 mole = 8 molecules
- Given that 1 mole (8 molecules) of H₂O weighs 180 g, the mass of one molecule is

$$\frac{180}{8} = 22.5 \text{ g}$$

Correct Answer: B

Question 20



So we want to produce 28g Fe

- (1) 1mol Fe = 56g => 0.5mol = 28g
- (2) We are going to find the moles Al needed = 0.5 mol
- (3) We are going to find the mass of Al needed = 0.5 mol x 27 = 13.5 g

Correct Answer: A

Question 21

We are going to determine:

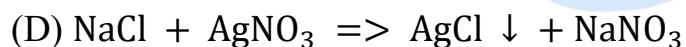
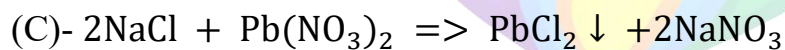
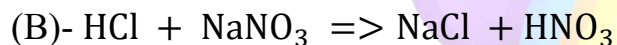
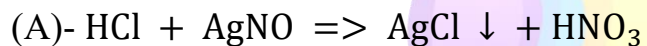
$$\% \text{ of Carbon} = \frac{\text{molar mass of Carbon in the molecule}}{\text{molar mass of the molecule}}$$

And now we are going to check the answers

A	B	C	D
$\% \text{ C} = \frac{12}{44} = 27,3 \%$	$\% \text{ C} = \frac{24}{30} = 80 \%$	$\% \text{ C} = \frac{72}{78} = 92,3 \%$	$\% \text{ C} = \frac{12}{16} = 75 \%$

Correct Answer: B

Question 22



Option B is the answer since every other option gives a precipitate.

Correct Answer: B

Question 23

The relationship between Celsius (°C) and Fahrenheit (°F) is given by the formula:

$$F = \frac{9}{5}C + 32$$

So, by using the formula, we can find that the answer is c.

Correct Answer: C

Question 24

The dilution formula relates the concentration and volume of a solution before and after dilution:

$$C_1V_1 = C_2V_2$$

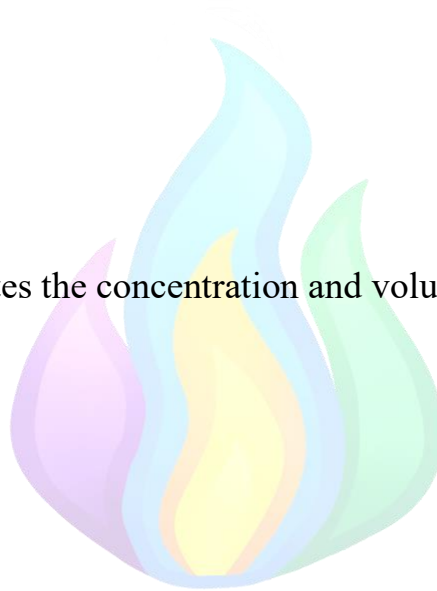
By using the formula

$$C_1V_1 = C_2V_2$$

$$(11)V_1 = 6(750)$$

$$V_1 = \frac{4500}{11} = 409 \text{ mL} = 0.409 \text{ L}$$

Correct Answer: B



Question 25

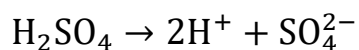
We know the formula for a mixture.

$$C_1V_1 + C_2V_2 = C_TV_T$$

$$0.25(0.1) + 0.2(0.05) = 0.45C_T$$

$$C_T = \mathbf{0.078\ M\ H_2SO_4}$$

H_2SO_4 fully ionizes, releasing $2\ H^+$ per molecule.

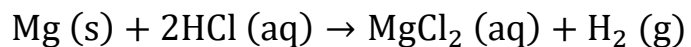


1M	2M
0.078	$\chi = \mathbf{0.156\ M}$

Correct Answer: B



Question 26



The gas produced in the reaction was 12,5 L in STP so

Step 1

1 mol H_2	22.4 L
$x = \mathbf{0.56\ mol\ H_2}$	12.5

Step 2

1 mol Mg	24g
0.56 mol	$x = \mathbf{13.4g}$

Correct Answer: A

Question 27

→ **X = $\text{CH}_3\text{CH}_2\text{CH}_3$ (Propane)**

- A small hydrocarbon (C_3H_8)
- **Propane is a gas at room temperature** due to weak van der Waals forces.

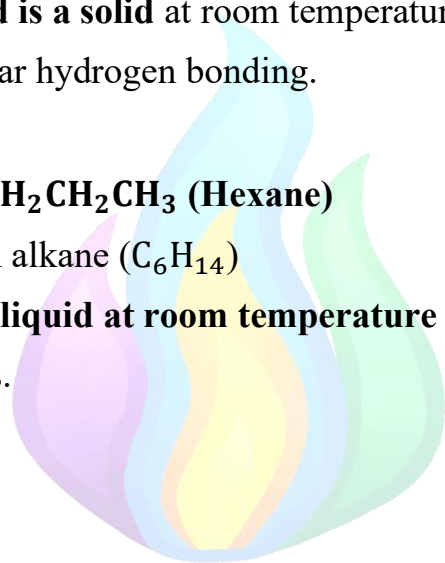
→ **Y = $\text{C}_6\text{H}_5\text{COOH}$ (benzoic acid)**

- A carboxylic acid with a benzene ring
- **Benzoic acid is a solid at room temperature** due to strong intermolecular hydrogen bonding.

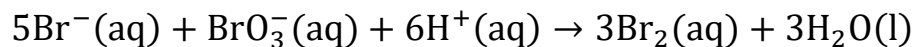
→ **Z = $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ (Hexane)**

- A six-carbon alkane (C_6H_{14})
- **Hexane is a liquid at room temperature** due to moderate van der Waals forces.

Correct Answer: - B



Question 28



The reaction involves

- 1 mole of BrO_3^- producing 3 moles of Br_2 .
- The given rate of disappearance of BrO_3^- is 0.10 Ms^{-1} .
- From the stoichiometry of the reaction:
1 mol of BrO_3^- produces 3 mol of Br_2

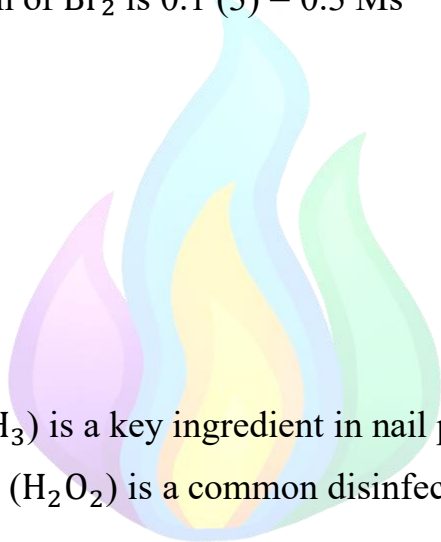
Thus, the rate of formation of Br_2 is $0.1 (3) = 0.3 \text{ Ms}^{-1}$

Correct Answer: C

Question 29

- Acetone (CH_3COCH_3) is a key ingredient in nail polish remover.
- Hydrogen Peroxide (H_2O_2) is a common disinfectant.

Correct Answer: C



Question 30



→ Given: 2 moles of CH_3OH and 1 mole of CO .

→ Since CH_3OH and CO react in a 1:1 ratio, the limiting reagent is CO (1 mole) because CH_3OH is in excess.

→ Molar mass of $\text{CH}_3\text{COOH} = 60$

Since 1 mole of CO should produce 1 mole of CH_3COOH , the theoretical yield is 60 g

→ The actual yield is 40.0 g, so the percentage yield is $(40/60) 100 = 66.6\%$.

Correct Answer: C

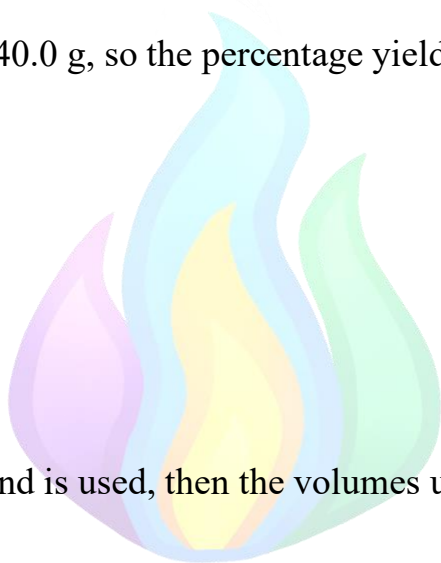
Question 31

If mass x of each compound is used, then the volumes used will be

$$V_{\text{tot}} = \frac{x}{7} + \frac{x}{3} = \frac{10x}{21}$$

$$\rho = \frac{2x}{\left(\frac{10x}{21}\right)} = 4.2$$

Correct Answer: C



Question 32

For this problem, we use the ideal transformer formula:

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

As we know that:

$$V_1 = 240 \text{ V}$$

$$V_2 = 12 \text{ V}$$

$$N_1 = 600$$

We solve:

$$N_2 = \frac{V_2}{V_1} \cdot N_1 = 30$$

Correct Answer: B



Question 33

To start, we should use the given data to find the resistance of the lamps. For this, it's important to use Ohm's Law:

$$I = \frac{V}{R}$$

In the first case, we know that

$$I = 10\text{mA} = 0.01 \text{ A}$$

As the two batteries are connected in series:

$$V = V_1 + V_2 = 4V$$

And as for the resistances:

$$R_{t1} = 2R \text{ (in series)}$$

And:

$$R_{t2} = \frac{1}{2} R \text{ (in parallel)}$$

Therefore, we get

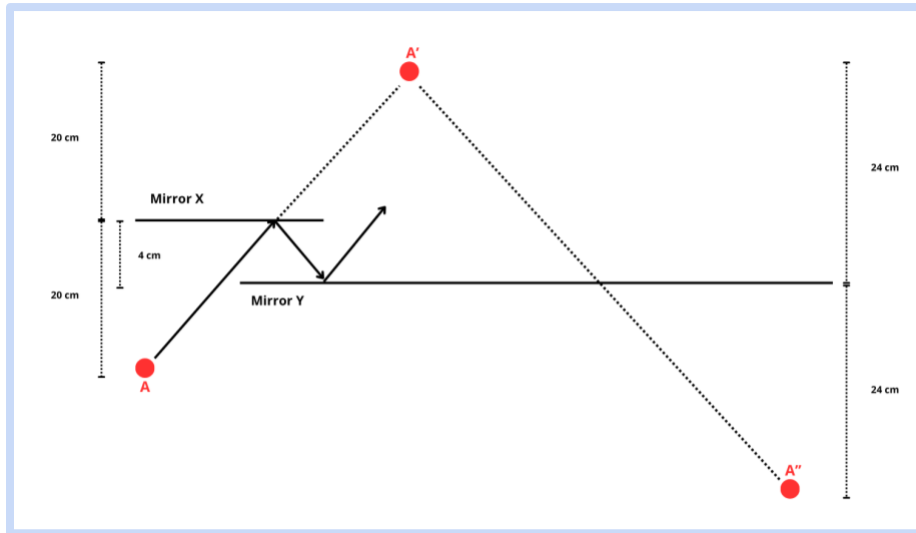
$$I_1 = \frac{4V}{2R} = 0.01 \text{ A}$$

$$I_2 = \frac{4V}{\frac{1}{2}R} = 4I_1 = 40 \text{ mA}$$

Correct Answer: C

Question 34

By drawing the virtual images and their respective distances:



The distance from mirror X to the final virtual image, A'' is going to be

$$d_{X-A''} = 24 \text{ cm} + 4 \text{ cm} = 28 \text{ cm}$$

So, the correct answer is 28 cm.

Correct Answer: C

Question 35

To solve this, we must use the specific heat formula applied to the transferred heat from the cup to the wind.

$$Q = mc\Delta T$$

As energy must be conserved, the Q that the wind wins must be the $-Q$ that the cup loses.

$$Q_{\text{wind}} = -Q_{\text{cup}}$$

By calling the flow rate v :

$$Q_{\text{wind}} = c_1 m(25 - 20) = c_1 vt(25 - 20)$$

$$Q_{\text{cup}} = c_2 m(40 - 90)$$

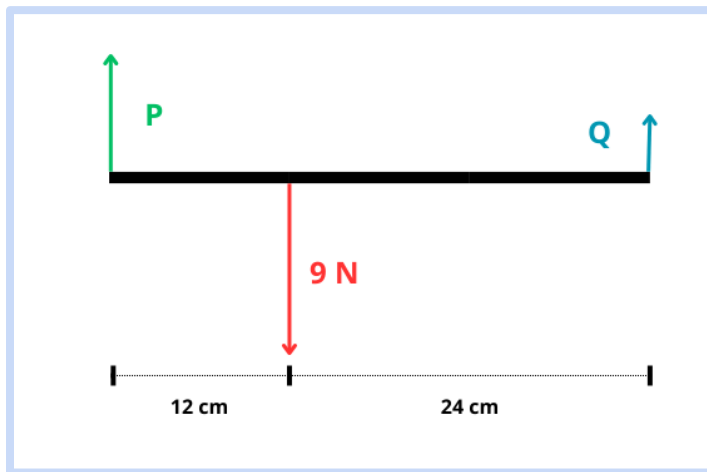
We finally solve for t :

$$t = \frac{c_2 m(90-40)}{c_1 v(25-20)} = 80\text{s}$$

Correct Answer: D

Question 36

As we know the rod is suspended, it's in equilibrium both in terms of forces and torques.



Forces:

IMP: We don't consider the weight of the rod because we're told that it's light, and we're not given any additional data; therefore, we don't have to take it into account.

$$P + Q = 9\text{N}$$

Torques:

We take the point where the force is applied as the fulcrum.

$$P (12\text{cm}) = Q (24\text{cm})$$

By solving the equations, we conclude that:

$$P = 6\text{N}$$

$$Q = 3\text{N}$$

Correct Answer: C

Question 37

The two cars will meet when they're at the same x position, and as we know that the travelled distance can be expressed in the graph as the area below the curve, we're looking for a time when:

$$\frac{1}{2}(v_A - 5)t + 5t = \frac{1}{2}v_B t$$

$$v_A + 5 = v_B$$

Then we find each velocity as a function of t to find the exact point where this happens. By using the linear equation formula:

$$y = mx + c$$

We get that:

$$v_A = \frac{1}{12}t + 5$$

$$v_B = \frac{1}{3}t$$

By replacing in the equation we had, we get to the conclusion that:

$$t = 40 \text{ min}$$

Correct Answer: C

Question 38

IMPORTANT NOTATION: A word is missing → “atmospheric pressure is equal to the pressure exerted by the liquid 10m down”

For this problem, we’ll use Boyle’s Law, which states that:

$$P_1 V_1 = P_2 V_2$$

And to find the pressure, we’ll use the Hydrostatic Pressure Equation:

$$P = P_0 + \rho gh$$

With P_0 being the atmospheric pressure. And ρgh being the pressure exerted by the liquid.

As we know that the atmospheric pressure is equal to the liquid pressure 10m down, we can equal:

$$P_0 = 10\rho g = P_1$$

And,

$$P_2 = P_0 + 20\rho g = 30\rho g$$

Finally, we get that,

$$V_1 = 3V_2$$

And as we know that $V_1 = 6 \text{ cm}^3$, we get that $V_2 = 18 \text{ cm}^3$

Correct Answer: D

Question 39

All answers are correct because:

- F.1: In case you made the same mistake as me, the fact that the incident angle is higher than the critical angle does not affect because total reflection only happens when $n_1 > n_2$. And, as air's refraction index is higher than the refraction index of glass, Figure 1 is correct.
- F2: In this case, total reflection is happening because $42^\circ > \theta_c$. Figure 2 is also correct.
- F3: Total reflection is happening inside the block because $45^\circ > \theta_c$. And when it leaves the block, the incident angle equals 0, which means that Figure 3 is correct.

Correct Answer: D



Question 40

For this problem, we'll use the fact that energy must be conserved to write the equation:

$$U_P + K_P = U_Q + K_Q + W$$

By the given data, we know that

$$U_Q = U_P - 50$$

$$K_Q = 5$$

$$W = 10$$

And by replacing, we get that

$$K_Q = 45 \text{ kJ}$$

Correct Answer: B

Question 41

By using Newton's second law:

$$\sum F = ma$$

On the first block:

$$T = 5 \cdot 2 = 10 \text{ N}$$

$$F - T = 3 \cdot 2 = 6 \text{ N}$$

And by solving the equation's system:

$$F = 16 \text{ N}$$

$$T = 10 \text{ N}$$

Correct Answer: B



Question 42

For this problem, we must use the formula

$$P = \frac{W}{t} = \frac{F \cdot d}{t}$$

By replacing the given data:

$$P = \frac{600 \text{ N} \cdot 15 \text{ m}}{30 \text{ s}}$$

$$P = 3000 \text{ W}$$

Correct Answer: C

Question 43

As the switch is closed, current flows through the circuit, and the iron nail becomes magnetized. By the right-hand rule, we know that the North Pole of the nail becomes the upper part, and as the pivoted light rod has some metallic material on the left side, point P goes up. As the current is still running, it stays that way.

Correct Answer: C

Question 44



We know that:

$$\rho = \frac{m}{v} = \frac{(180 - 100) \text{ g}}{(40 - 30) \text{ cm}^3}$$

$$\rho = 8 \text{ g} \cdot \text{cm}^{-3}$$

Correct Answer: C

Question 45

By using the lens equation:

$$\frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}$$

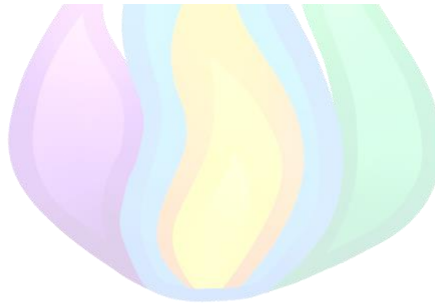
And replacing that:

$$d_i = 2f$$

We get that

$$d_0 = 2f = \text{POINT C}$$

Correct Answer: C



Question 46

In questions like this one, we must change all terms into powers of 2.

$$\text{right - hand side} = 2^{x-1} * 8 = 2^{x-1} * 2^3 = 2^{x+2}$$

$$\text{left - hand side} = 2^{2x-2}$$

Since the left - hand side equals the right side, $2^{x+2} = 2^{2x-2}$

$$\text{So, } x + 2 = 2x - 2$$

$$X = 4$$

Correct Answer: D

Question 47

First, let's rearrange the equation.

$$R = \frac{(S + T)^2}{3}P$$

$$\frac{3R}{P} = (S + T)^2$$

Now, let's take the square root of both sides.

$$\sqrt{\frac{3R}{P}} = S + T$$

$$\sqrt{\frac{3R}{P}} - T = S$$

Correct Answer: C



Question 48

The gradient, $m = \frac{\Delta y}{\Delta x} = \frac{-1 - 3}{5 - (-1)} = \frac{-4}{6} = \frac{-2}{3}$

Now, we will find the equation of the graph.

When $x = 5$, $y = -1$.

$$y - y' = m(x - x')$$

$$y - (-1) = \frac{-2}{3}(x - 5)$$

$$y = \frac{-2}{3}x + \frac{7}{3}$$

The y-intercept is $+7/3$.

Correct Answer: D

Question 49

$$\frac{4x + 3}{2} - \frac{2x - 1}{3} = \frac{3(4x + 3) - 2(2x - 1)}{6}$$

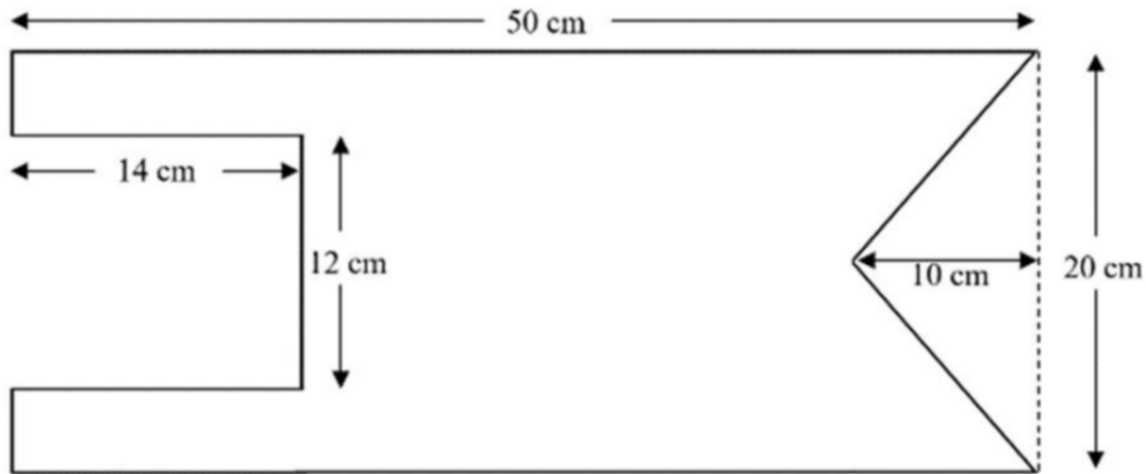
$$= \frac{12x + 9 - 4x + 2}{6} = \frac{8x + 11}{6} = 3$$

$$8x + 11 = 18$$

$$x = \frac{7}{8}$$

Correct Answer: D

Question 50



In a question like this, the easiest thing to do is to find the area of the big rectangle and subtract the area of the rectangle and the area of the triangle.

$$\text{Area of the big rectangle} = \text{width} * \text{length} = 50\text{cm} * 20\text{cm} = 1000\text{cm}^2$$

$$\text{Area of the small rectangle} = \text{width} * \text{length} = 14\text{cm} * 12\text{cm} = 168\text{cm}^2$$

$$\text{Area of the triangle} = \frac{1}{2} * \text{base} * \text{height} = \frac{1}{2} * 10\text{cm} * 20\text{cm} = 100\text{cm}^2$$

$$\text{Area of the shape} = 1000\text{cm}^2 - 168\text{cm}^2 - 100\text{cm}^2 = 732\text{cm}^2$$

Correct Answer: C